Asmit – 2401072 – T21 - ASDD

AWS CI/CD PIPELINE TO HOST A STATIC WEBSITE

Using GitHub, AWS CodePipeline, AWS CodeBuild and Amazon S3

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1. Aim

To design and implement an automated CI/CD pipeline on Amazon Web Services (AWS) for deploying and hosting a static website using GitHub, AWS CodePipeline, AWS CodeBuild and Amazon S3.

The pipeline automatically detects changes pushed to the GitHub repository, builds the website using AWS CodeBuild, and deploys the updated website to an Amazon S3 bucket.

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2. Objectives

The objectives of this experiment are:

1. To create a static website using HTML, CSS and JavaScript.
2. To store the website source code in a GitHub repository.
3. To configure an AWS CodeBuild project.
4. To configure AWS CodePipeline for continuous integration and deployment.
5. To connect GitHub with AWS CodePipeline.
6. To automatically build the website whenever changes are pushed.
7. To deploy the website to an Amazon S3 bucket.
8. To enable static website hosting on Amazon S3.
9. To verify automatic deployment after modifying the source code.

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3. Technologies Used

Technology	Purpose
GitHub	Source code repository
AWS CodePipeline	CI/CD pipeline orchestration
AWS CodeBuild	Build and package the website
Amazon S3	Website hosting and deployment
AWS IAM	Access and permissions management
Amazon CloudWatch	Build log monitoring

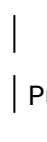
Technology	Purpose
HTML	Website structure
CSS	Website styling
JavaScript	Website functionality

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4. System Architecture

The CI/CD pipeline implemented in this experiment follows the workflow below:

Developer



GitHub Repository



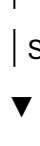
AWS CodePipeline



AWS CodeBuild



Amazon S3



Deployed Website

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Pipeline Flow

Code Change

↓

GitHub Push

↓

CodePipeline Trigger

↓

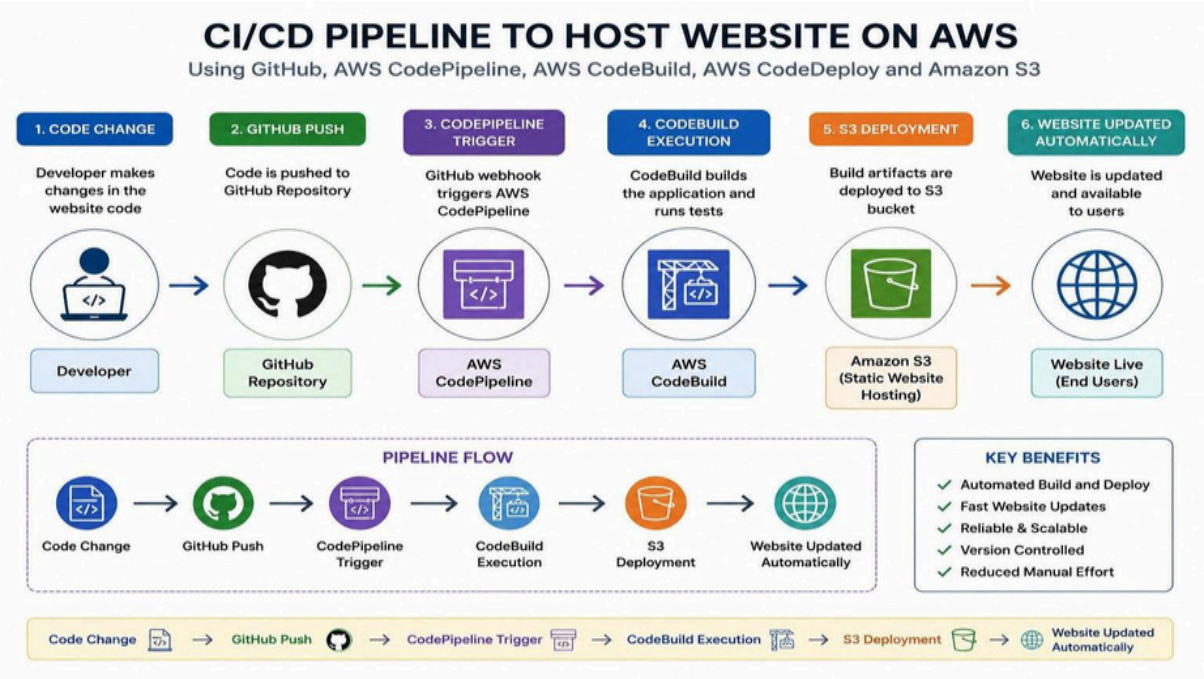
CodeBuild Execution

↓

S3 Deployment

↓

Website Updated Automatically



5. Project Structure

The static website used for this experiment contains the following files:

```
my-cicd-project/
│
├── index.html
│
├── css/
│   └── style.css
│
├── js/
│   └── script.js
│
├── images/
│
└── buildspec.yml
```

Description

- index.html – Main webpage.
- css/style.css – Styling for the website.
- js/script.js – JavaScript functionality.
- images/ – Directory for website images.
- buildspec.yml – Instructions used by AWS CodeBuild during the build process.

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6. Buildspec File

The buildspec.yml file is placed in the root directory of the GitHub repository.

```
version: 0.2


phases:
  install:
    commands:
      - echo "Starting AWS CodeBuild..."
```

```
 pre_build:
  commands:
   - echo "Preparing website files..."

 build:
  commands:
   - echo "Building static website..."
   - echo "Build completed successfully."

 post_build:
  commands:
   - echo "Preparing deployment artifacts..."

artifacts:
 files:
  - '**/*'
```

The buildspec file defines the commands executed by AWS CodeBuild and specifies the files that are included in the build artifact.

---

7. Procedure

Step 1: Create the Website

Create a static website using HTML, CSS and JavaScript.

The project should contain:

index.html

css/style.css

js/script.js

images/

buildspec.yml

The website can contain a homepage describing the AWS CI/CD pipeline.

Step 2: Create a GitHub Repository

Create a new repository on GitHub.

Example repository name:

my-cicd-project

Upload or push the following files:

index.html

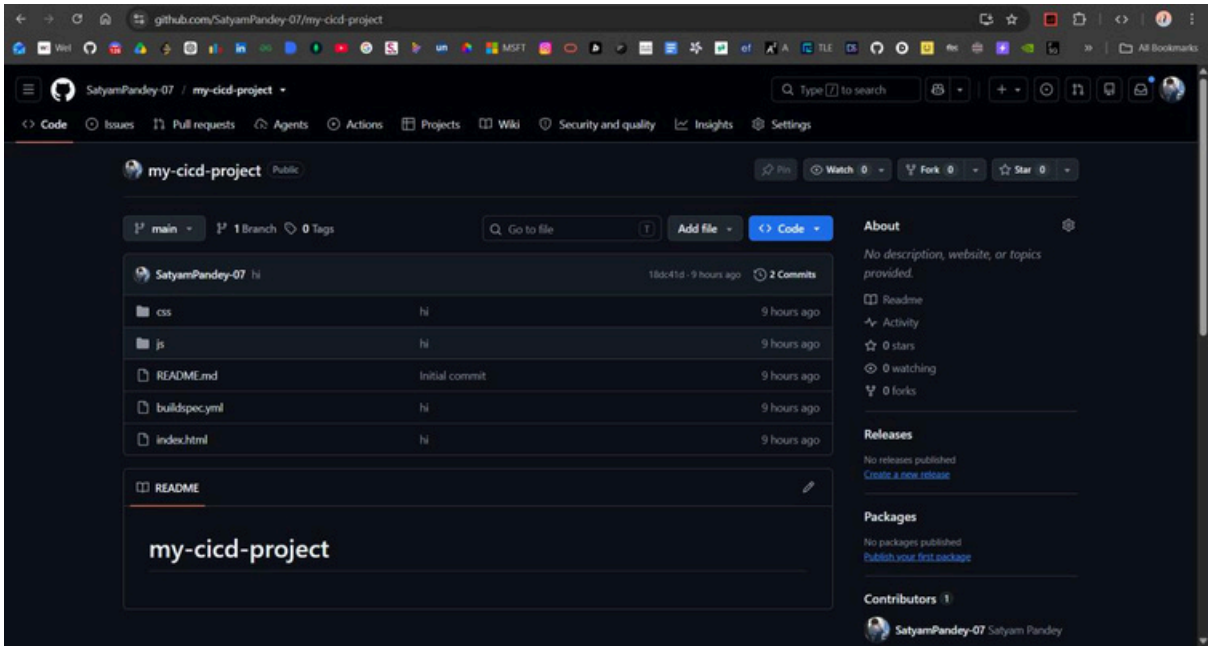
buildspec.yml

css/

js/

images/

The GitHub repository acts as the source code repository for the CI/CD pipeline.



Step 3: Push the Project to GitHub

Initialize Git in the project directory and push the source code to GitHub.

Example commands:

git init

git add .

git commit -m "Add AWS CI/CD website"

git branch -M main

git remote add origin https://github.com/YOUR_USERNAME/my-cicd-project.git

git push -u origin main

After pushing, verify that all project files are visible in the GitHub repository.

---

8. Create AWS CodeBuild Project

Step 4: Open AWS CodeBuild

Log in to the AWS Management Console.

Search for:

CodeBuild

Open AWS CodeBuild and select:

Build projects → Create build project

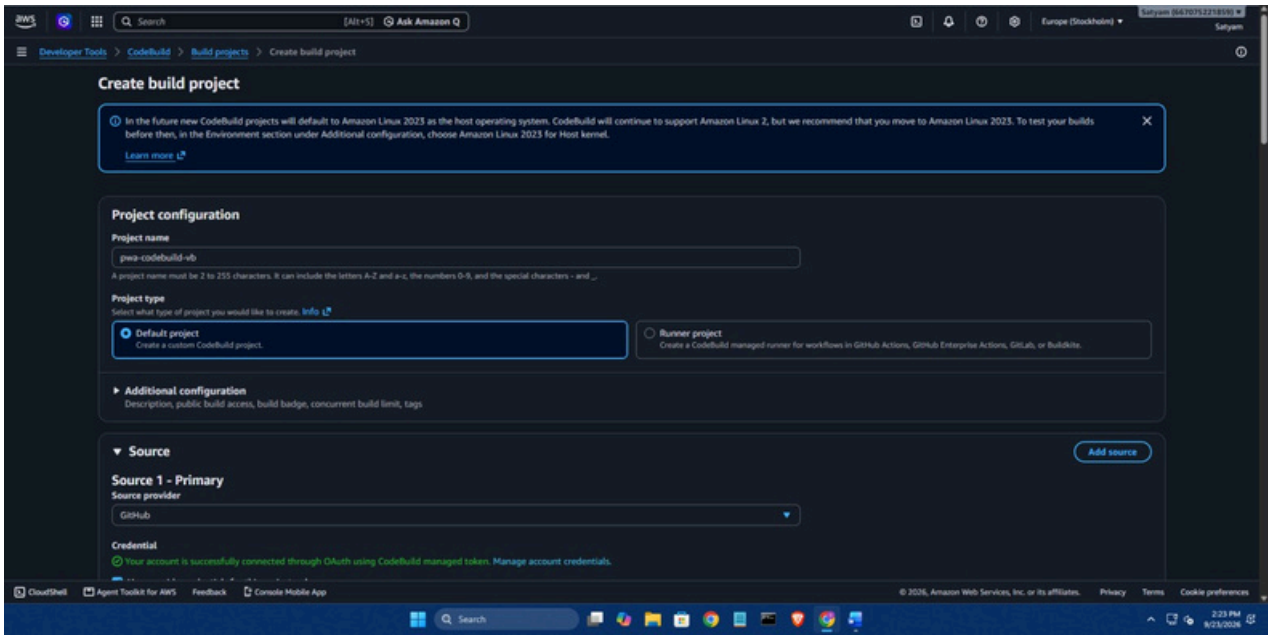
Step 5: Configure Project Name

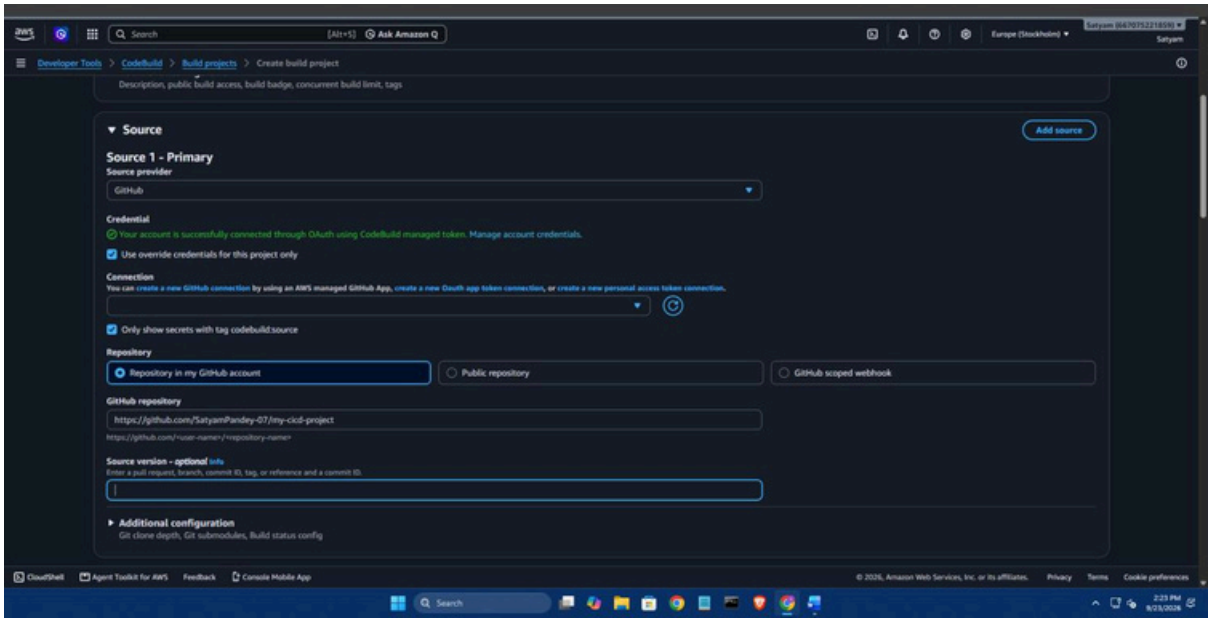
Enter the project name.

Example:

my-cicd-project

The project name uniquely identifies the CodeBuild project.





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9. Configure Source

Step 6: Select Source Provider

Under the Source section:

Select:

Source provider → GitHub

Connect the GitHub account using the available GitHub authentication/OAuth process.

Authorize AWS CodeBuild when prompted.

The GitHub repository will then become available for selection.

Step 7: Select Repository

Select:

Repository in my GitHub account

Choose the repository:

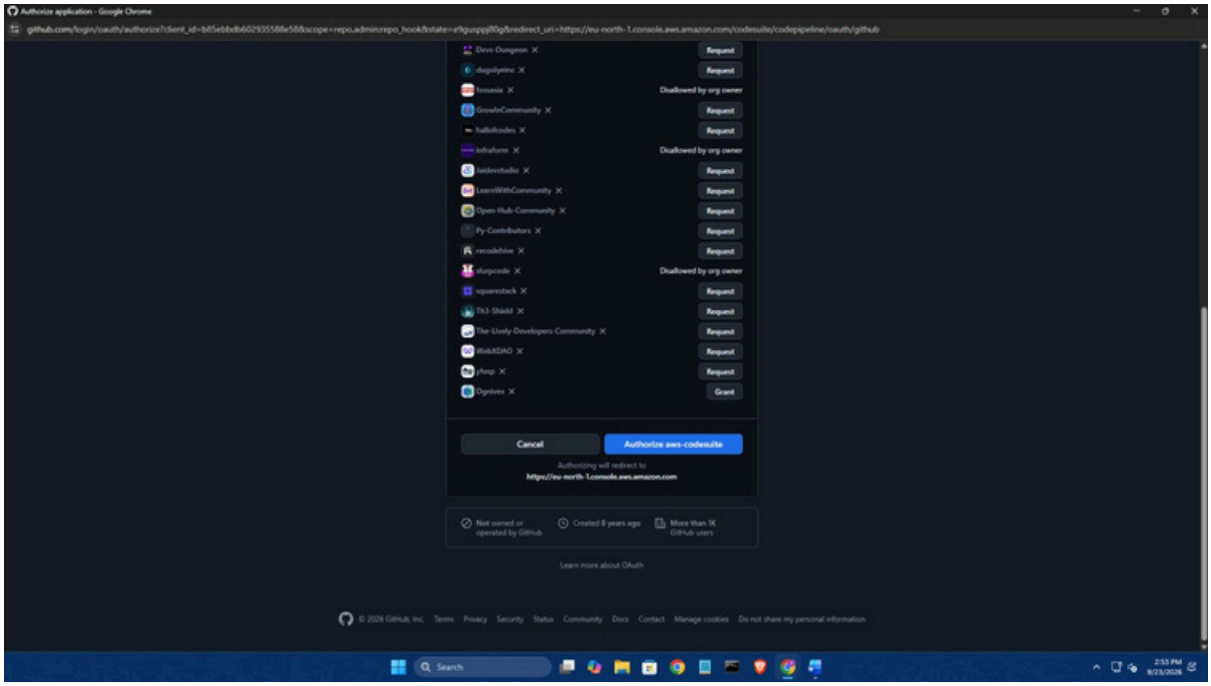
my-cicd-project

Select the branch containing the source code.

For example:

main

The selected GitHub repository becomes the primary source for the CodeBuild project.



---

10. Configure Environment

Step 8: Configure Build Environment

Scroll to the Environment section.

Select:

Environment image:

Managed image

Select:

Running mode:

Instance

Select:

Operating system:

Amazon Linux

For the runtime, use:

Base

Use the AWS-managed CodeBuild image.

Step 9: Configure Service Role

Under Service role, select:

New service role

AWS will automatically create a service role for CodeBuild.

Example:

codebuild-my-cicd-project-service-role

The service role allows CodeBuild to interact with the AWS resources required during the build.

---

11. Configure Buildspec

Step 10: Select Buildspec File

Scroll to the Buildspec section.

Select:

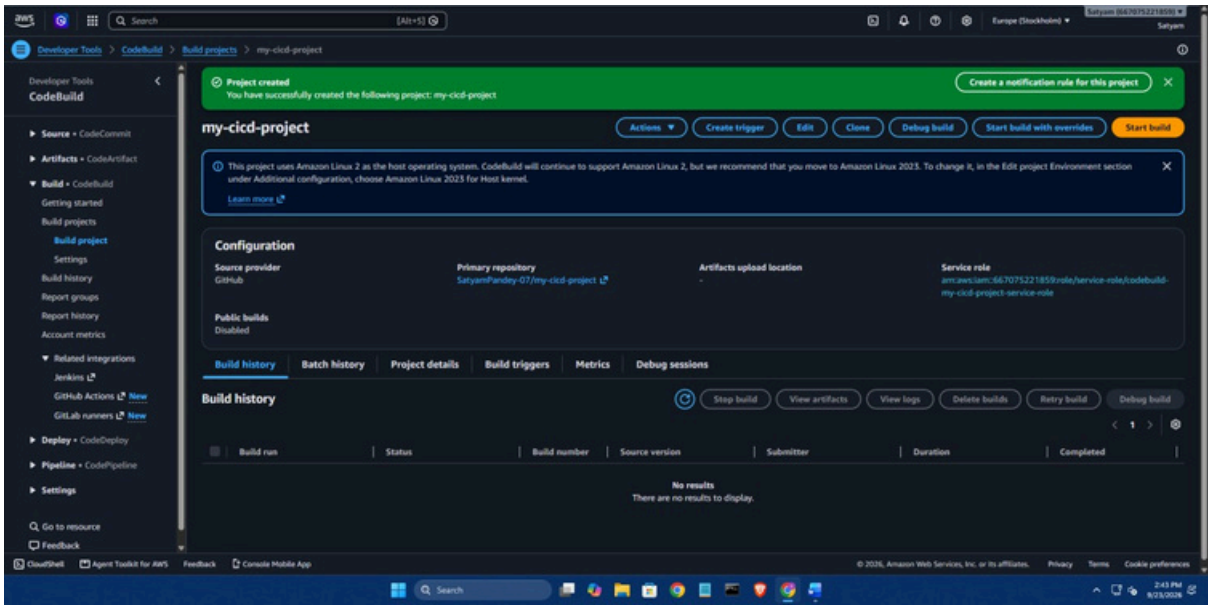
Use a buildspec file

Use:

buildspec.yml

The file is already present in the root directory of the GitHub repository.

AWS CodeBuild will read the instructions from this file during execution.



---

12. Configure Artifacts

Step 11: Configure Artifacts

For this CodeBuild project, select:

No artifacts

The pipeline will handle the artifacts during the CodePipeline workflow.

The buildspec file still specifies the files that can be used as build output when CodeBuild is invoked by CodePipeline.

---

13. Configure Logs

Step 12: Enable CloudWatch Logs

Under the Logs section, enable:

CloudWatch logs

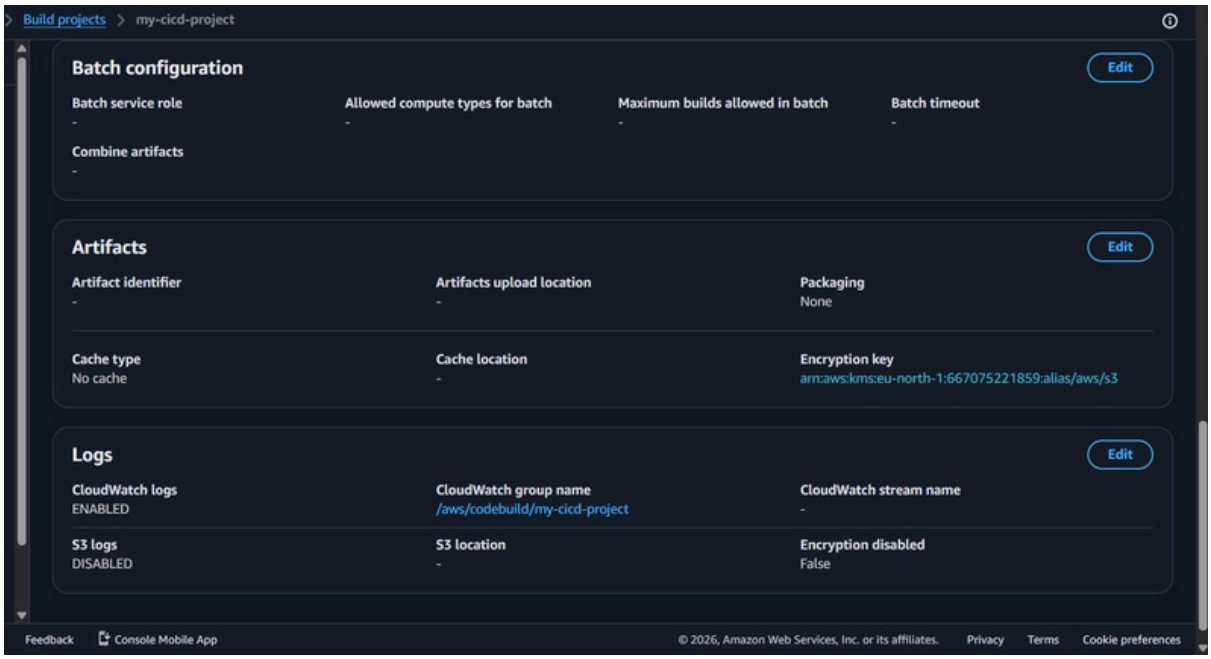
The default log group can be used, for example:

/aws/codebuild/my-cicd-project

S3 build logs are not required for this experiment.

After reviewing the configuration, click:

Create build project



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14. Verify CodeBuild Project

Step 13: Verify Project Creation

After creating the project, AWS CodeBuild displays the project details.

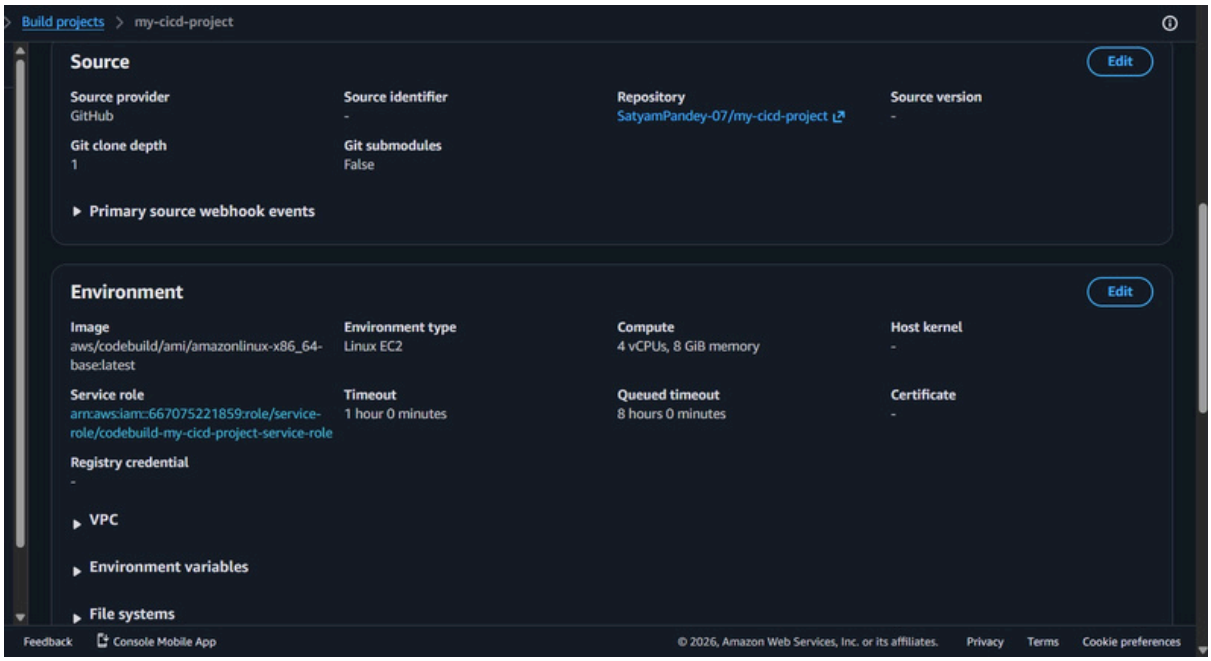
Verify that the project has been successfully created.

The CodeBuild project should show:

Source: GitHub

Environment: Managed Image

Buildspec: buildspec.yml



---

15. Create AWS CodePipeline

Step 14: Open CodePipeline

Search AWS for:

CodePipeline

Open:

AWS CodePipeline → Pipelines

Select:

Create pipeline

In the newer AWS interface, select:

Build custom pipeline

This allows the individual Source, Build and Deploy stages to be configured manually.

16. Configure Pipeline Settings

Step 15: Configure Pipeline

Enter a pipeline name.

Example:

my-cicd-pipeline

Select:

New service role

AWS will create the required CodePipeline service role.

The pipeline stores and passes artifacts between its different stages.

---

17. Configure Artifact Store

Step 16: Configure Artifact Store

Under advanced settings, configure the artifact store.

Use an Amazon S3 bucket for storing CodePipeline artifacts.

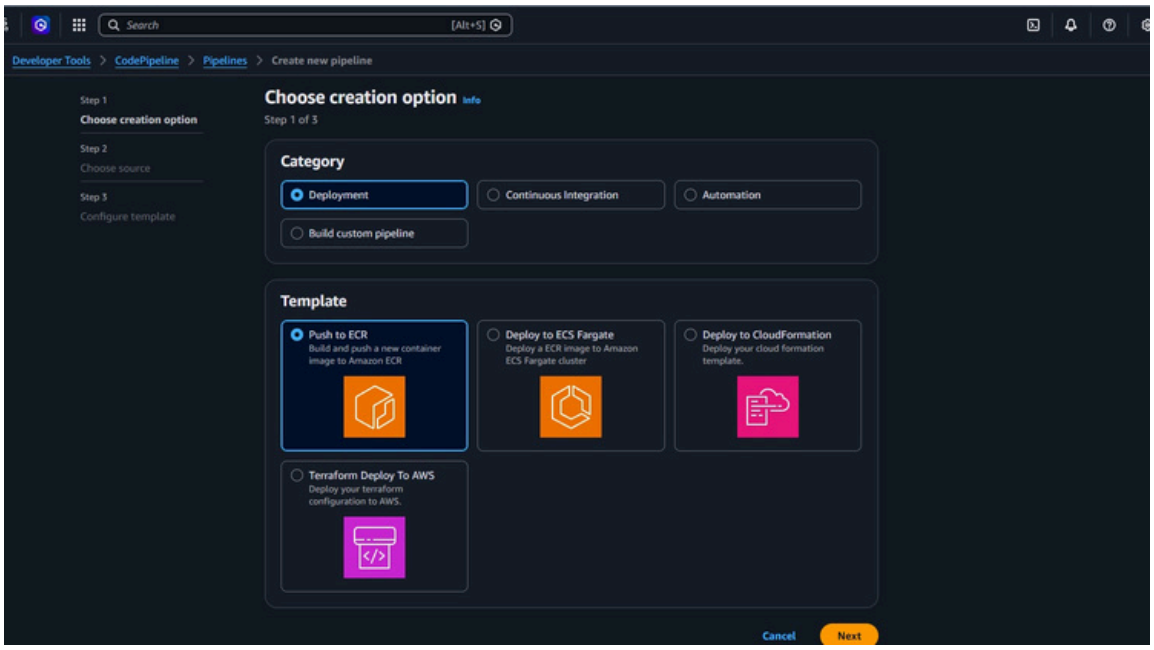
Example:

store-build-artifact-vb

Use the default AWS-managed encryption key unless a different key is required by the assignment.

Click:

Next



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18. Configure Source Stage

Step 17: Add GitHub Source

For the Source stage:

Select:

Source provider → GitHub

Connect/authenticate your GitHub account.

Select:

Repository:

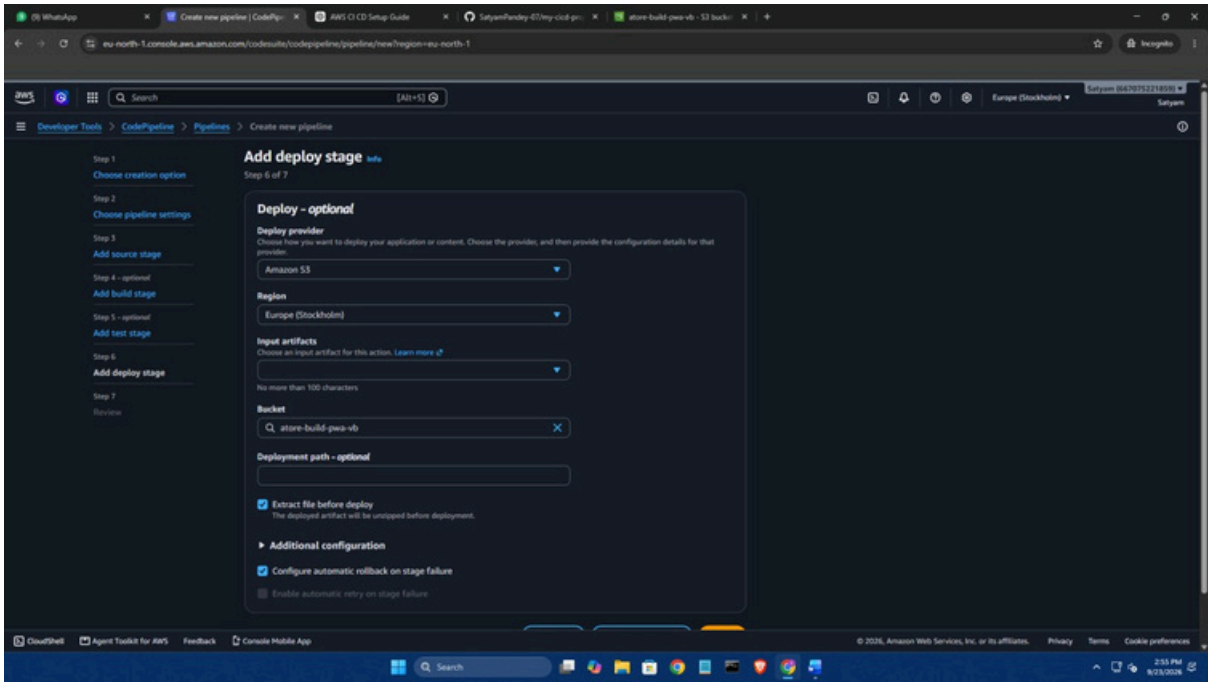
my-cicd-project

Select the branch:

main

Enable automatic change detection if available.

This allows CodePipeline to automatically detect new changes pushed to the repository.



---

19. Configure Build Stage

Step 18: Add Build Stage

Select:

Build provider

→ Other build providers

→ AWS CodeBuild

Select the CodeBuild project created earlier:

my-cicd-project

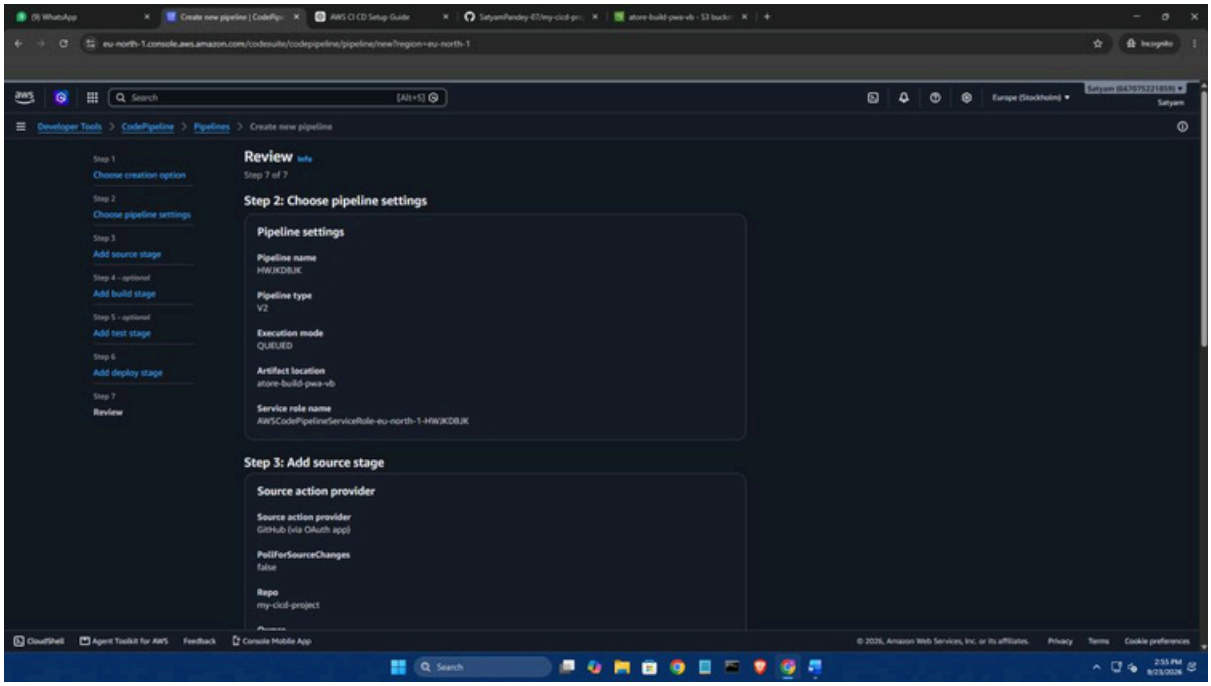
The SourceArtifact generated from the GitHub source stage is provided as the input to CodeBuild.

Use:

Build type: Single build

Click:

Next



---

20. Configure Deploy Stage

Step 19: Add Deployment Stage

For the deployment stage, select:

Deploy provider → Amazon S3

Select the S3 deployment bucket.

Example:

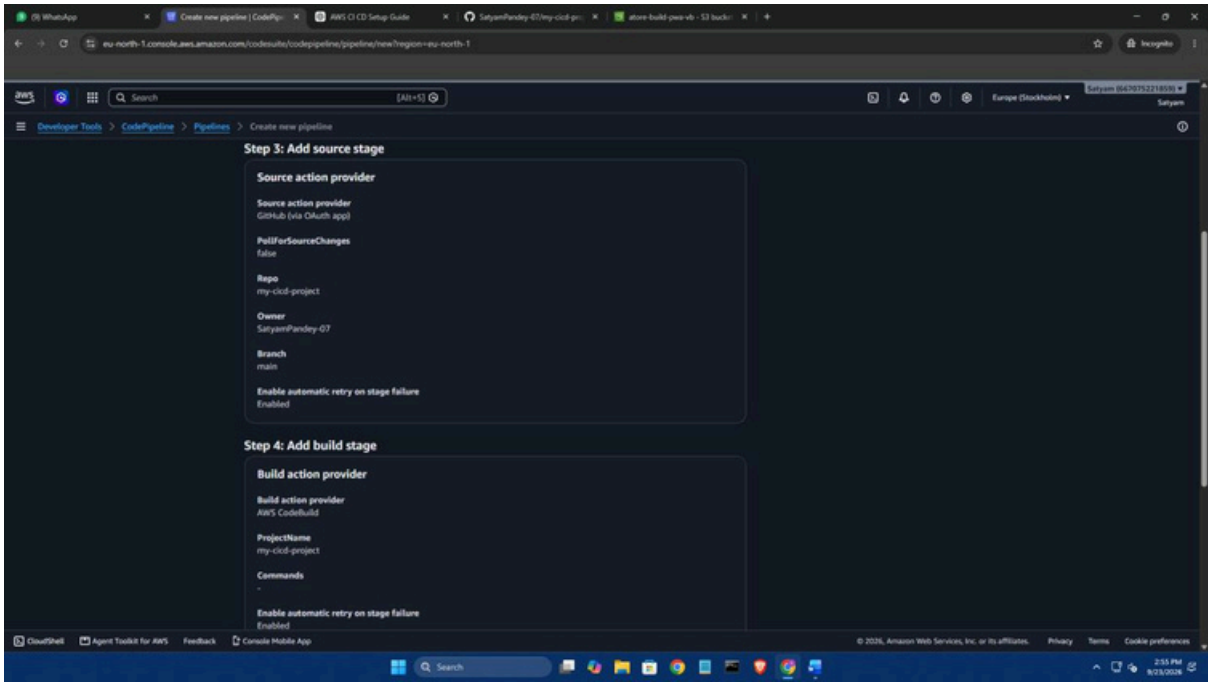
deployment-bucket-pwa-vb

Select the input artifact produced by the build stage.

Enable the option to extract the artifact before deployment if available.

Click:

Next



---

21. Create Pipeline

Step 20: Review and Create Pipeline

Review the complete pipeline configuration.

The pipeline should contain three major stages:

SOURCE

↓

BUILD

↓

DEPLOY

Where:

GitHub

↓

AWS CodePipeline

↓

AWS CodeBuild

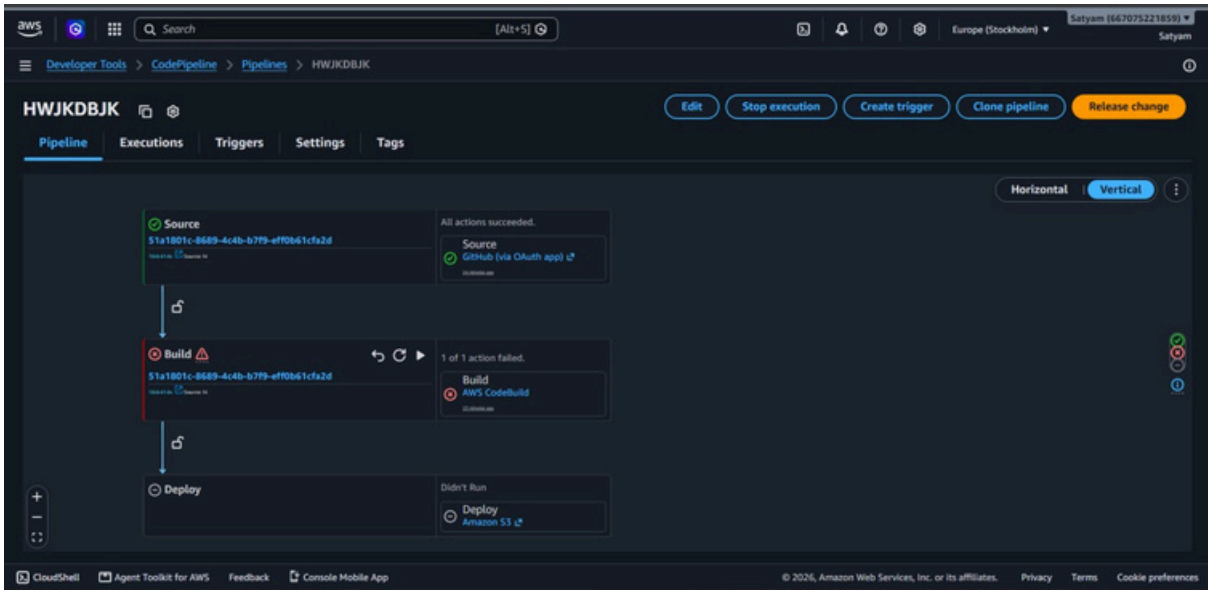
↓

Amazon S3

Click:

Create pipeline

AWS will create and execute the pipeline.



---

22. Handle Build Permission Issues

Step 21: Verify Build Execution

If the build fails because of insufficient permissions, open:

AWS CodeBuild

→ Build project

→ Build details

→ Service role

The service role opens in AWS IAM.

Open:

Permissions

and select:

Add permissions

→ Attach policies

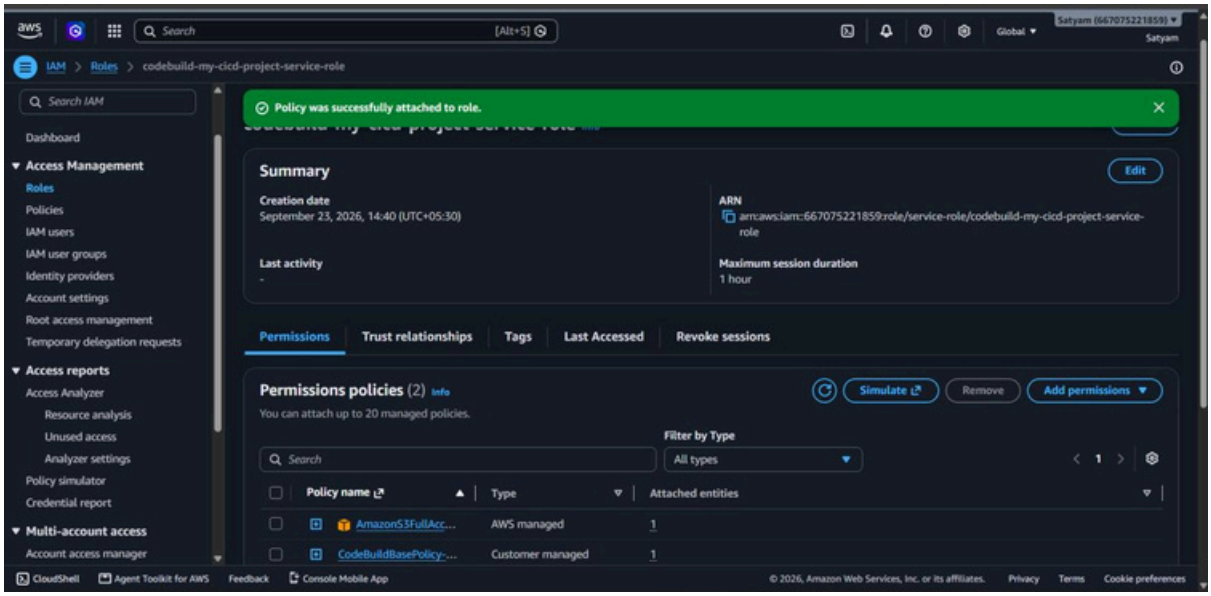
Search for the required S3 permission.

The reference procedure uses:

AmazonS3FullAccess

and attaches it to the CodeBuild service role.

Note: For a production system, permissions should normally be restricted to only the required S3 actions/resources rather than granting full S3 access.



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23. Retry Pipeline

Step 22: Retry Pipeline

Return to:

AWS CodePipeline

→ my-cicd-pipeline

Retry the failed execution.

Wait for the pipeline to complete.

The expected result is:

Source ✓

Build ✓

Deploy ✓

The pipeline should now execute successfully.

---

24. Configure Amazon S3 Static Website Hosting

Step 23: Open S3 Deployment Bucket

Search AWS for:

S3

Open the deployment bucket.

Go to:

Properties

Scroll to:

Static website hosting

Click:

Edit

Enable:

Static website hosting

Select:

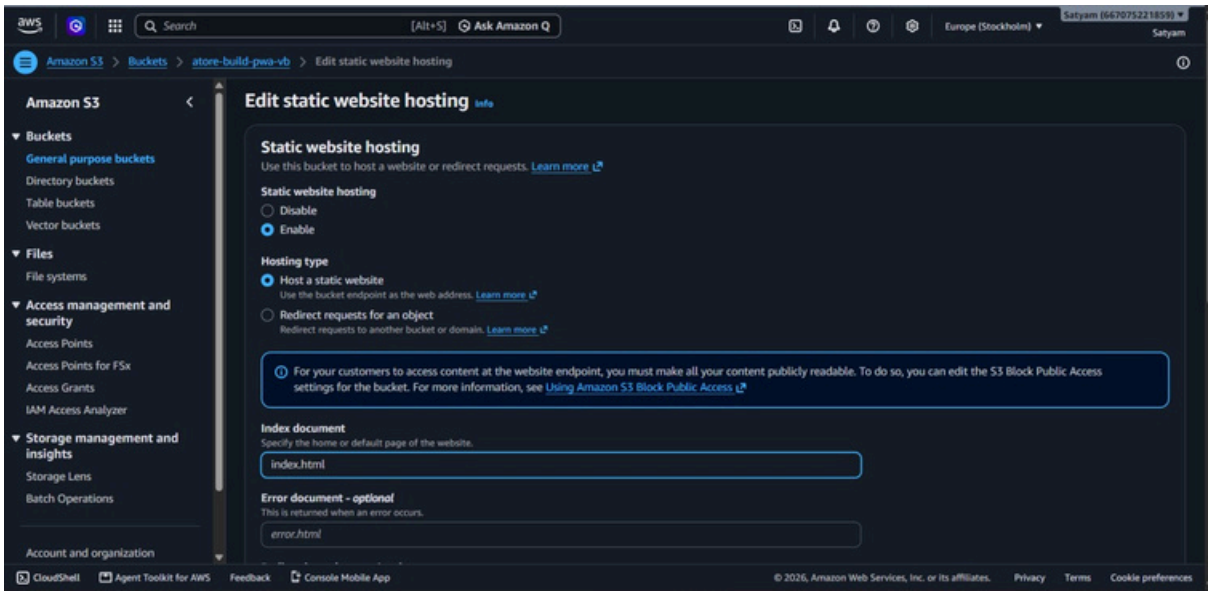
Host a static website

Set the index document to:

index.html

Save the changes.

The reference document follows the same process of enabling static website hosting and specifying index.html.



---

25. Configure S3 Bucket Policy

Step 24: Edit Bucket Policy

Go to:

S3

→ Deployment Bucket

→ Permissions

→ Bucket policy

Add a policy that permits public GetObject access if your assignment specifically requires a publicly accessible S3 website.

Use your actual bucket name in the resource ARN.

Example:

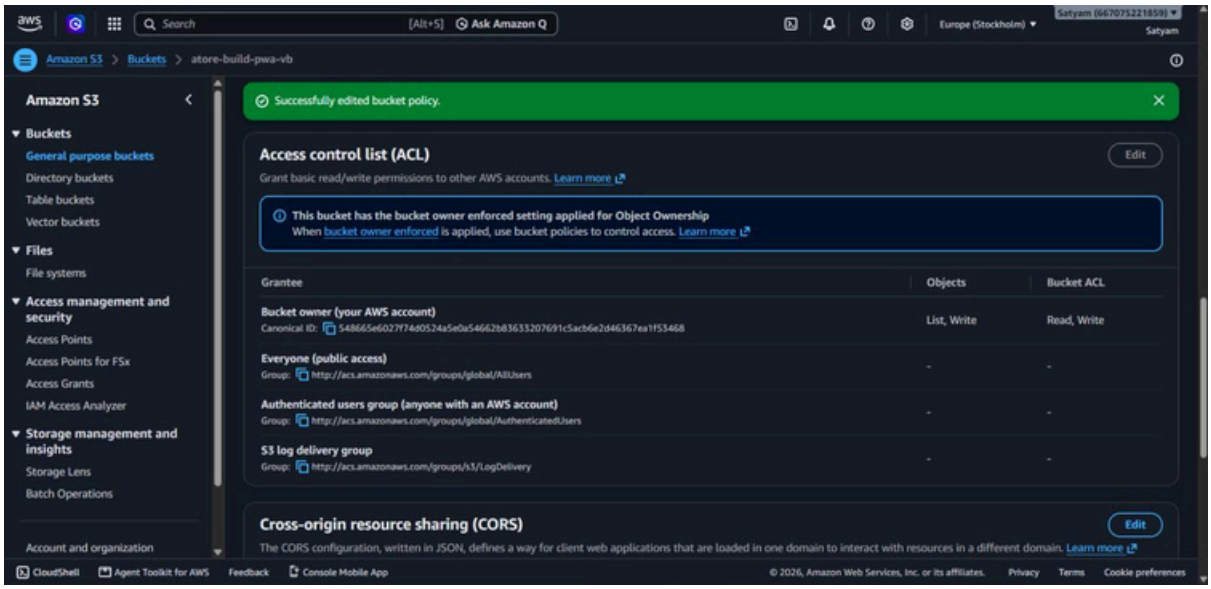
```
{
 "Version": "2012-10-17",
 "Statement": [
  {
   "Sid": "PublicReadGetObject",
   "Effect": "Allow",
   "Principal": "*",
   "Action": "s3:GetObject",
   "Resource": "arn:aws:s3:::YOUR-BUCKET-NAME/*"
  }
 ]
}
```

Replace:

YOUR-BUCKET-NAME

with the actual S3 bucket name.

The reference document uses the same public s3:GetObject pattern.



26. Open the Deployed Website

Step 25: Access Website

Return to:

S3

→ Deployment Bucket

→ Properties

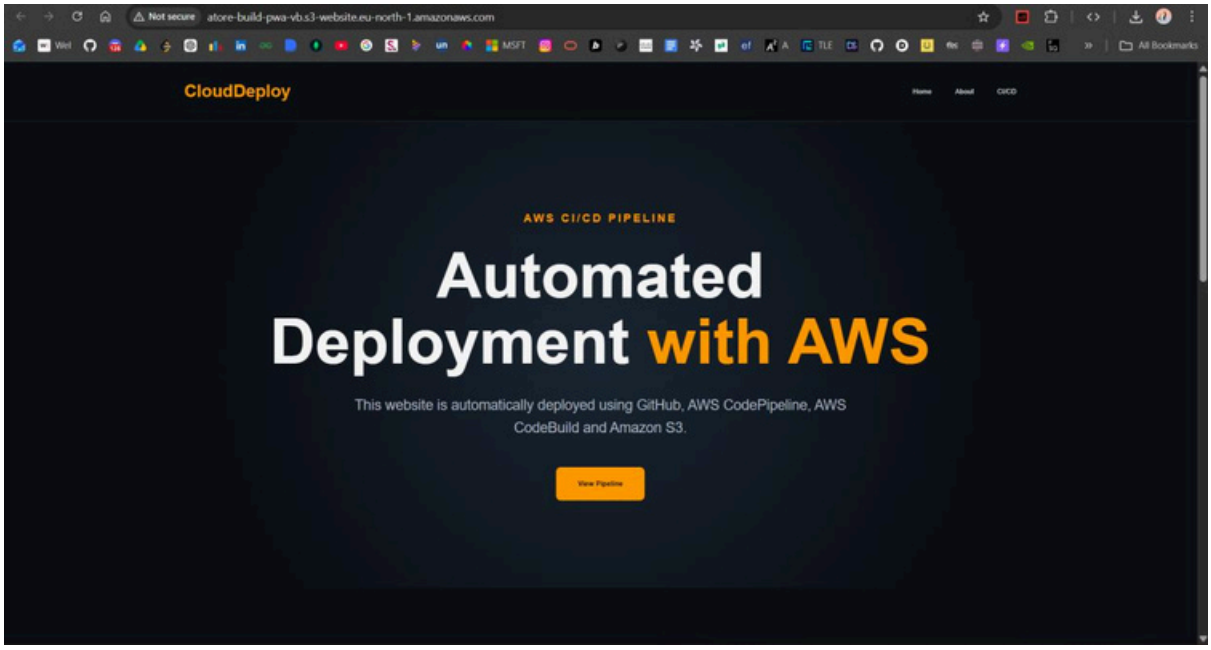
Scroll to:

Static website hosting

AWS displays the Bucket website endpoint.

Click the website endpoint.

The deployed static website should open in the browser.



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27. Test Automatic Deployment

Step 26: Modify the Source Code

To verify that the CI/CD pipeline works automatically, make a small change to the website.

For example, modify the heading in index.html.

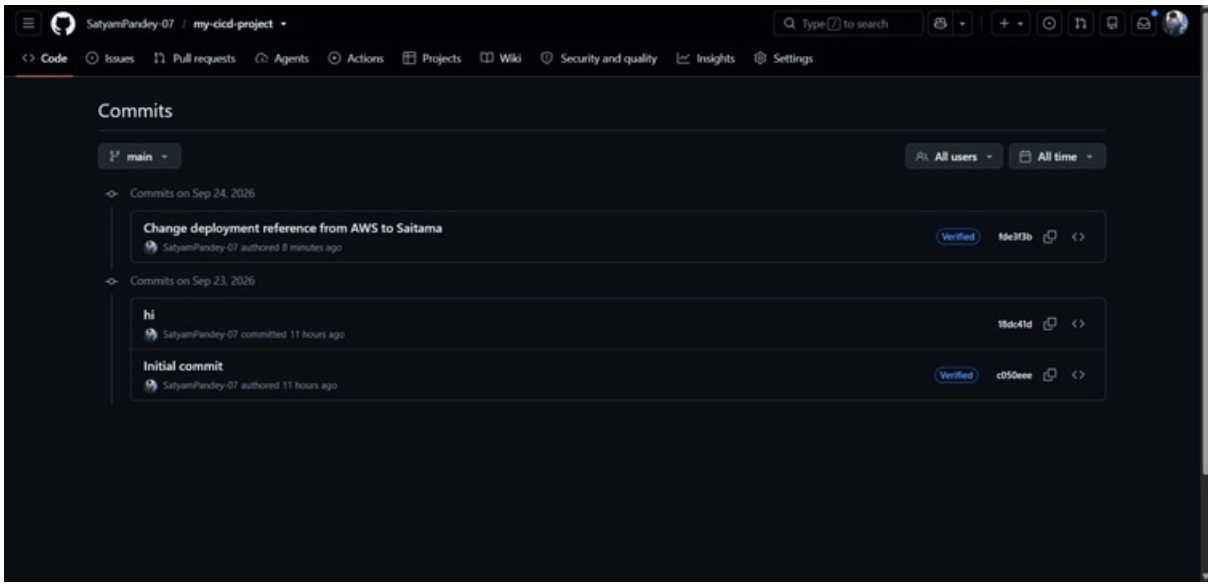
Before:

<h1>Automated Deployment with AWS</h1>

After:

<h1>Automated Deployment with Saitama</h1>

Commit and push the changes to GitHub.



---

28. Verify Automatic Pipeline Trigger

Step 27: Check CodePipeline

Open:

AWS CodePipeline

→ my-cicd-pipeline

A new pipeline execution should start after the GitHub change is detected.

The stages should execute in sequence:

GitHub

↓

Source

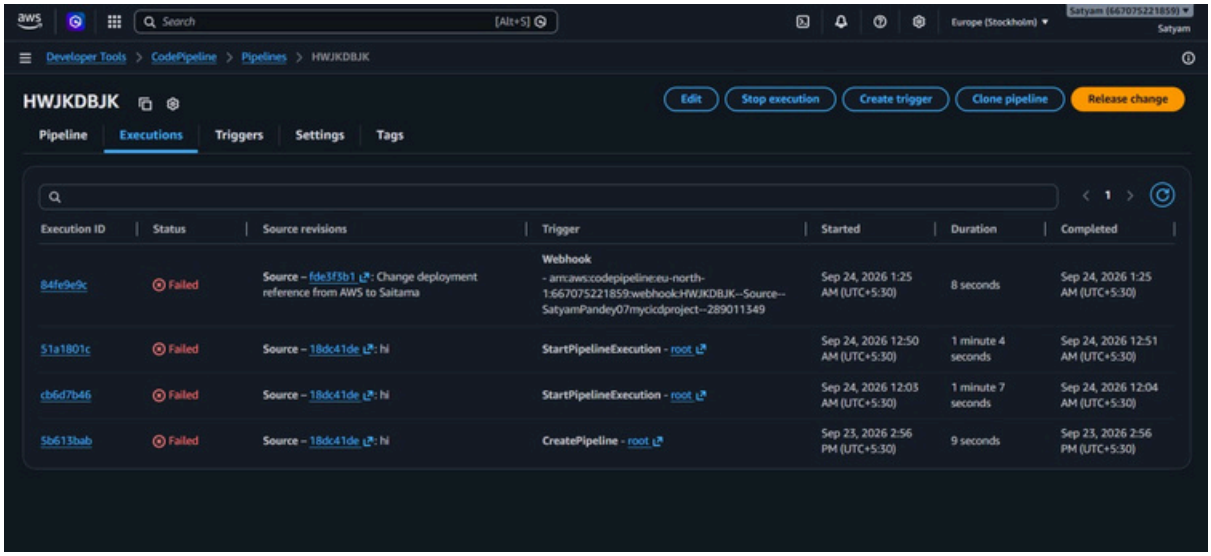
↓

Build

↓

Deploy

Wait until all stages become successful.



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29. Verify Updated Website

Step 28: Refresh the Website

Open the S3 static website endpoint again.

Refresh the page.

The changes made in GitHub should now be visible on the deployed website.

This verifies that the complete CI/CD workflow is functioning:

GitHub Change

↓

CodePipeline Trigger

↓

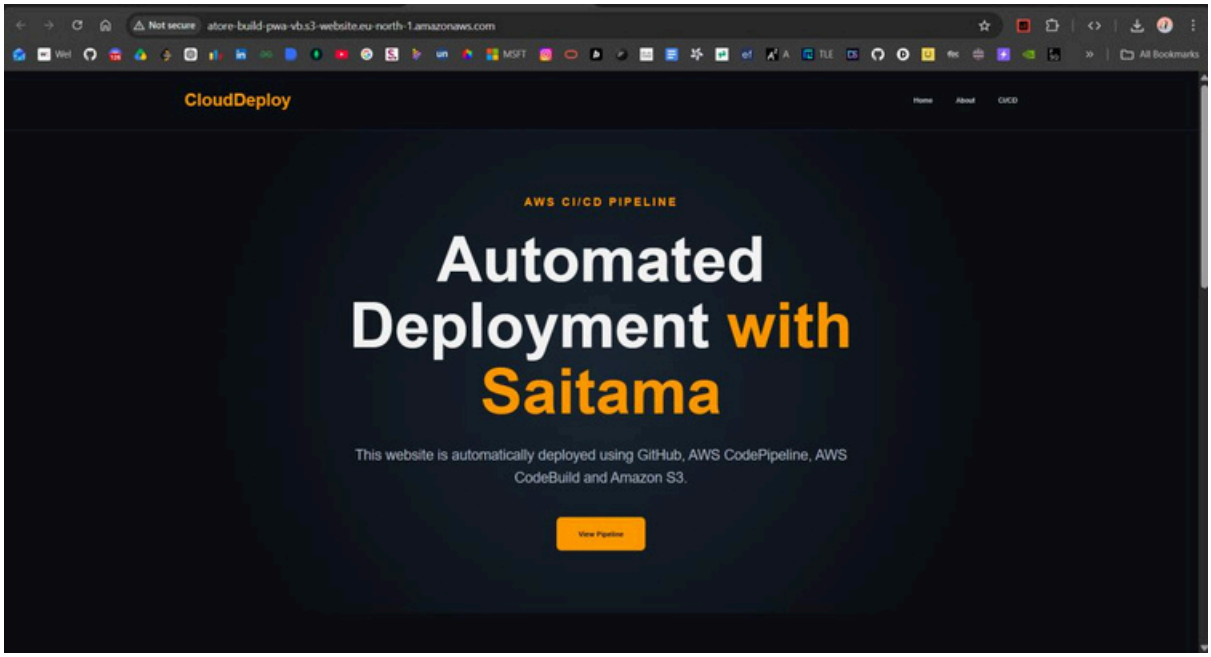
CodeBuild

↓

S3 Deployment

↓

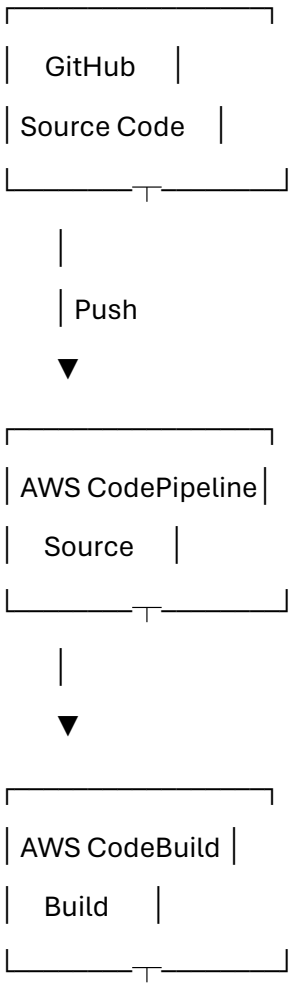
Updated Website

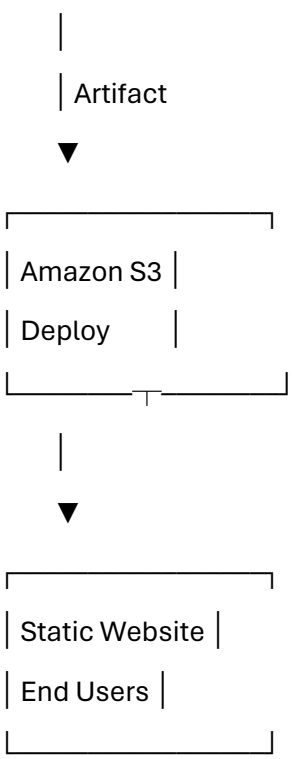


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30. Final CI/CD Workflow

The final implementation can be represented as:





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31. Result

The AWS CI/CD pipeline was successfully configured using GitHub, AWS CodePipeline, AWS CodeBuild and Amazon S3.

The source code was maintained in GitHub, automatically processed through AWS CodePipeline and CodeBuild, and deployed to an Amazon S3 static website hosting environment.

A subsequent change made to the GitHub repository triggered a new pipeline execution and updated the deployed website automatically.

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32. Conclusion

Thus, the AWS CI/CD Pipeline for automated static website deployment was successfully implemented.

Amazon S3

↓

Static Website

The implementation demonstrates how source-code changes can be automatically built and deployed without manually uploading updated website files to the hosting environment.
